

Abhinav Raj

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Profile Summary

Software engineer specializing in **AI-driven applications, backend development, and Prompt engineering**. Proficient in **Python, FastAPI, Docker, and PostgreSQL**, with experience in deploying AI models and optimizing scalable APIs. Successfully integrated **NudeNet AI for content moderation** in an e-commerce platform. Passionate about **data science, AI, and cloud computing**.

Education

B.Tech, Computer Science and Engineering

APJ Abdul Kalam Technological University

2020 – 2024

Experience

Backend Developer & AI Engineer — ESSBEE TechStrategies, Infopark Koratty

2024

- Designed and deployed **AI-based content moderation system (NudeNet)** for an e-commerce platform.
- Developed and optimized scalable **RESTful APIs** using **FastAPI**.
- Managed **Dockerized microservices**, improving system efficiency and reliability.
- Designed robust **PostgreSQL schemas** and implemented **Redis-based caching**.
- Conducted **unit testing and integration testing** to ensure API stability.
- Collaborated with frontend developers to ensure seamless backend integration.

CTO — Innovation and Entrepreneurship Development Cell (IEDC), SNMIMT

2022 – 2023

- Conducted **AI and cloud computing workshops** to guide students.
- Led multiple teams working on AI/ML-driven applications and product prototypes.

Technical Skills

Programming Languages: Python, C

Backend Technologies: FastAPI, Flask

Databases: PostgreSQL, MS SQL, Redis

AI & Data Science: PyTorch, TorchVision, TensorFlow, OpenCV, LangChain

AI APIs: OpenAI API, Gemini API, ChatGPT, RAG, Prompt Engineering

Cloud & DevOps: Docker, Kubernetes (basic)

Tools: Git, Docker, VS Code, PyCharm, Jira, CI/CD

Protocols: REST API, WebSockets, RTSP, HTTP

Message Queues: Celery, RabbitMQ

Projects

AI-Based Content Moderation (Python, FastAPI, NudeNet, Docker)

- Integrated **NudeNet AI model** for detecting explicit content in an e-commerce platform.
- Deployed the AI model in a **Dockerized environment**, ensuring scalability and efficiency.
- Designed an API-based solution for automated content filtering.

DeepFake Detection (Python, Flask, PyTorch, OpenCV)

- Developed an AI-powered system to **detect deepfake images and videos**.
- Leveraged **PyTorch and TorchVision** for training deep learning models.
- Integrated **CUDA acceleration for deep learning inference**, enhancing inference speed.

Prompty (Prompt Optimization, AI-driven User Profiling – GitHub)

- Developed a **privacy-first AI prompt optimizer** as a Chrome extension.
- Utilized **local LLMs** for real-time AI prompt enhancement across ChatGPT, Claude, and Poe.
- Implemented **Ollama-based model fine-tuning** for optimized prompt suggestions.

Hackathon Competition - GuruHacks

- Managed event logistics and operations as the **main coordinator**, managing event logistics and operations.
- Secured **sponsorships** and organized AI-focused coding challenges.

Professional Skills

- AI/ML Model Deployment and API Integration**
- Strong Problem-Solving and Algorithmic Thinking**
- Cloud & DevOps (Docker, Kubernetes, AWS Basics)**
- Effective Collaboration and Agile Development**
- Testing & Debugging (Unit Testing, Integration Testing)**